Lightweight Face Recognition for Edge Video Deployment via MagFace-Guided Knowledge Distillation and Track-Level Quality-Aware Retrieval

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Abstract

While modern deep face recognition models achieve strong accuracy, they are often too computationally expensive for edge deployment and lack the robust tracking, quality control, and retrieval mechanisms required for practical video scenarios. To address this, this work presents a lightweight video face recognition pipeline featuring a MobileNetV4-Conv-Medium student model distilled from a frozen MagFace iResNet-100 teacher. The system evaluates three training variants to balance clean accuracy and robustness: a clean knowledge distillation baseline, a spatial-KD variant with aggressive occlusion, and a Stochastic Weight Averaging stabilized variant designed for occluded faces. This distilled model is integrated into an end-to-end runtime pipeline that includes YOLO-based detection, alignment, tracking, liveness filtering, track-level pooling, and FAISS-based retrieval. Experimental results demonstrate that the clean KD baseline delivers the best low-FAR verification on clean faces, whereas the SWA-stabilized variant significantly improves robustness under lower-face occlusion, offering a practical trade-off for efficient edge-oriented face recognition. https://github.com/Prism190/VGU_FaceRec_Project

1. Introduction

Face recognition systems are increasingly deployed in real-world video scenarios such as access control, surveillance, and attendance monitoring. Although modern deep face recognition models achieve strong accuracy on standard benchmarks, they are often too computationally expensive for edge deployment and are usually evaluated under image-level settings. Practical video-based recognition requires not only an accurate embedding model, but also robust detection, alignment, tracking, liveness filtering, quality control, and identity retrieval.

This work presents a lightweight video face recognition pipeline based on a MobileNetV4-Conv-Medium student distilled from a frozen MagFace iResNet-100 teacher[12]. The student learns 512-dimensional face embeddings using a combination of MagFace classification loss and embedding-level knowledge distillation[7, 15]. To study the trade-off between clean accuracy and robustness, we evaluate three independently trained variants of the same teacher-student framework: a clean KD baseline, a spatial-KD variant with aggressive occlusion curriculum, and an SWA-stabilized variant designed for occluded faces[9]. These variants are not sequential training stages; each is trained from scratch under a different objective and augmentation policy.

The distilled model is integrated into an end-to-end runtime pipeline consisting of YOLO-based face detection, 5-point alignment, tracking, liveness filtering, MagFace magnitude-based quality gating, magnitude-weighted track pooling, and FAISS-based identity retrieval[5, 10, 17]. This design allows the system to exploit temporal information from video rather than relying on single-frame predictions. Experimental results show that the clean KD variant provides the best low-FAR verification performance on clean faces, while the SWA-based robust variant reduces degradation under lower-face occlusion. The proposed system therefore provides a practical trade-off between recognition accuracy, robustness, and deployment efficiency for edge-oriented face recognition.

2. Related Work

2.1. Face Recognition

Deep face recognition has achieved strong performance through discriminative embedding learning. Instead of directly comparing raw images, modern systems learn compact face embeddings in which samples from the same iden-

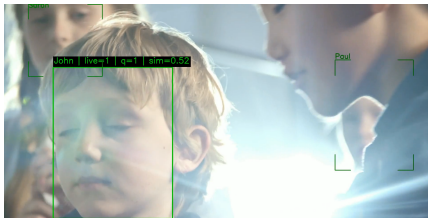
*Designed the Knowledge Distillation architecture, occlusion curriculum, and overall training pipeline

†Implemented cross-class NMS detection and tracking logic

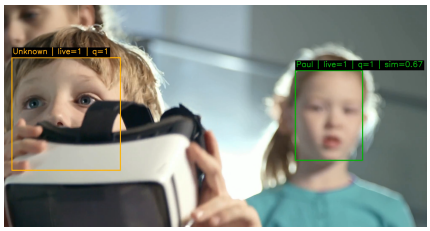
‡Implemented anti-spoofing confirmation cache, MagFace quality gating, and FAISS open-set clustering



(a) All visible subjects are correctly recognized by the system.



(b) The system maintains recognition under partial face occlusion.



(c) A heavily occluded face wearing a VR headset is rejected as unknown.

Figure 1. Qualitative examples of the proposed video face recognition pipeline. Green boxes indicate accepted recognized identities, while the yellow box indicates an unknown identity. The examples show that the pipeline can recognize visible faces, remain stable under partial occlusion, and reject a heavily occluded face when the visual evidence is insufficient for reliable identification.

tity are close together and samples from different identities are separated. Margin-based softmax losses have played an important role in this progress. CosFace introduces an additive cosine margin in the normalized feature space, while ArcFace applies an additive angular margin to enforce stronger inter-class separation with a clear geometric interpretation[20][4]. These methods have become standard objectives for face verification and identification.

MagFace further extends margin-based face recognition by associating feature magnitude with face quality[12]. In MagFace, high-quality face samples are encouraged to produce embeddings with larger magnitude, while low-quality or ambiguous samples tend to have smaller magnitude. This is useful not only during training but also during deployment, because embedding magnitude can be used as a quality indicator. Our system follows this idea by using a MagFace iResNet-100 model as the teacher and by using embedding magnitude for quality filtering and track-level pooling in the video pipeline.

2.2. Lightweight Face Recognition and Edge Deployment

Although large face recognition models can achieve high accuracy, they are often too computationally expensive for edge devices or real-time video applications. This has motivated the development of lightweight architectures such as MobileFaceNets and MobileNet-based models. These networks reduce parameter count and computational cost while attempting to preserve recognition accuracy[2].

MobileNetV4 continues this direction by providing efficient building blocks designed for mobile and edge hardware[16]. In this work, MobileNetV4-Conv-Medium is used as the student backbone. However, training a compact model only with classification supervision may not be sufficient to match the representation quality of larger models. Therefore, our approach combines the efficiency of MobileNetV4 with knowledge distillation from a stronger MagFace teacher.

2.3. Knowledge Distillation for Face Embeddings

Knowledge distillation transfers knowledge from a large teacher model to a smaller student model. In the original formulation, the student learns from the softened output distribution of the teacher. For face recognition, however, the final embedding space is more important than the class probability distribution, especially because recognition is usually performed through similarity comparison rather than direct classification.

Therefore, distillation for face recognition can be applied at the embedding level, where the student is encouraged to match the teacher's feature representation. Some works also explore relational distillation, where pairwise distances or angular relations between samples are transferred. In our system, the main distillation signal is embedding-level knowledge distillation from a frozen MagFace iResNet-100 teacher to a MobileNetV4 student. The student is also trained with a MagFace classification head, allowing it to learn both supervised identity separation and teacher-guided embedding structure[15][12][16].

2.4. Occlusion Robustness and Face Quality

Real-world face recognition often suffers from partial occlusion, pose variation, blur, illumination changes, and low-quality frames. Occlusion is especially challenging because important facial regions may be missing, while the model may still produce confident but unreliable embeddings. This issue becomes more serious in video scenarios, where only some frames may contain a clear frontal face.

To address this, many systems use data augmentation or quality-aware filtering. In our work, the training variants explore different occlusion strategies, including lower-face masking, Gaussian blur, and motion blur. The robust variant uses a milder occlusion curriculum combined with stochastic weight averaging to improve stability under masked or partially occluded conditions[9]. At inference time, embedding magnitude is used as a quality signal, so low-quality observations can be filtered before being added to the track-level representation.

2.5. Video Face Recognition Pipelines

A practical video face recognition system requires more than a face embedding model. It must first detect faces, align them, extract embeddings, associate detections across frames, filter unreliable observations, and finally retrieve the closest identity from a gallery. Face alignment is particularly important because poorly aligned crops can significantly reduce verification accuracy. Landmark-based alignment, such as 5-point affine alignment, is commonly used to normalize face images before feature extraction.

Tracking is also important in video-based recognition. Methods such as DeepSORT and BoT-SORT maintain identity consistency across frames by combining motion and appearance cues[22][1]. In our system, tracking is used not only to reduce identity flickering but also to collect multiple embeddings from the same person over time. Instead of making a recognition decision from a single frame, the system performs track-level pooling, which makes the final identity decision more stable.

2.6. Retrieval, Open-Set Recognition, and Unknown Grouping

After extracting a face embedding, recognition is usually performed by comparing it against a gallery of known identities. For small galleries, direct cosine similarity is sufficient, but larger galleries benefit from approximate nearest-neighbor search. FAISS is commonly used for efficient vector retrieval and is suitable for large-scale embedding search[5].

Real deployment also requires open-set behavior, where the system must decide whether a face belongs to a known identity or should be treated as unknown. Our pipeline uses threshold-based retrieval with a top-1 score and top-1/top-2 margin check to reduce false matches. If a face cannot be matched to a known identity, the system can group unknown tracks based on embedding similarity. This allows repeated appearances of the same unknown person to be clustered and reviewed later.

2.7. Face Anti-Spoofing and Liveness Detection

Face recognition systems used for authentication or access control are vulnerable to presentation attacks such as printed photos, replayed videos, or screen-based attacks. Therefore, liveness detection is an important component of a complete deployment pipeline. However, face anti-spoofing is a difficult problem because performance depends heavily on the attack type, camera domain, and availability of real presentation-attack data.

In our system, liveness detection is treated as a gate before embeddings are accepted into the track buffer. Only observations that pass both liveness filtering and quality filtering are used for track-level pooling and identity retrieval. This design reduces the risk that spoofed or low-quality frames corrupt the recognition decision, while keeping the anti-spoofing component modular so that stronger PAD models can be integrated later[13].

3. Method

We propose an edge-oriented video face recognition system built around a lightweight student face encoder. The runtime pipeline consists of face detection, alignment, embedding extraction, tracking, liveness filtering, quality gating, track-level pooling, and identity retrieval. The main encoder is a MobileNetV4-Conv-Medium student distilled from a frozen MagFace iResNet-100 teacher. MobileNetV4 is selected for efficient mobile and edge inference [16], while MagFace is used because its feature magnitude is associated with face quality [12]. This allows the embedding norm to be used both as a training signal and as a quality cue during video inference.

3.1. Teacher-Student Architecture

The teacher is a frozen MagFace iResNet-100 model that outputs 512-dimensional face embeddings. The student is a smaller MobileNetV4-Conv-Medium model with the same 512-dimensional embedding size. The student consists of a MobileNetV4 backbone, a projection layer, and batch normalization. For variants that use spatial distillation, an additional 1×1 projection head aligns the student spatial feature map with the teacher feature map.

During training, the teacher parameters are fixed. The teacher receives inputs in the range expected by the MagFace checkpoint, while the student receives normalized face crops. This design allows the student to learn a compact embedding space while preserving the representation structure of the stronger teacher.

3.2. Training Objective

The student is trained with supervised identity classification and teacher-guided embedding distillation. The overall loss is:

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{MagFace}} + \lambda_{\text{kd}}\mathcal{L}_{\text{KD}} + \lambda_{\text{spatial}}\mathcal{L}_{\text{spatial}}. \quad (1)$$

Here, $\mathcal{L}_{\text{MagFace}}$ is the MagFace classification loss, $\mathcal{L}_{\text{KD}}$ is the embedding-level distillation loss, and $\mathcal{L}_{\text{spatial}}$ is an optional spatial feature matching loss. ArcFace introduced additive angular margin learning for discriminative face embeddings [4], while MagFace extends margin-based face recognition by linking feature magnitude with face quality [12]. The main distillation term in this work is mean squared error between student and teacher embeddings, following the general idea of knowledge distillation from a large teacher to a compact student [7][15].

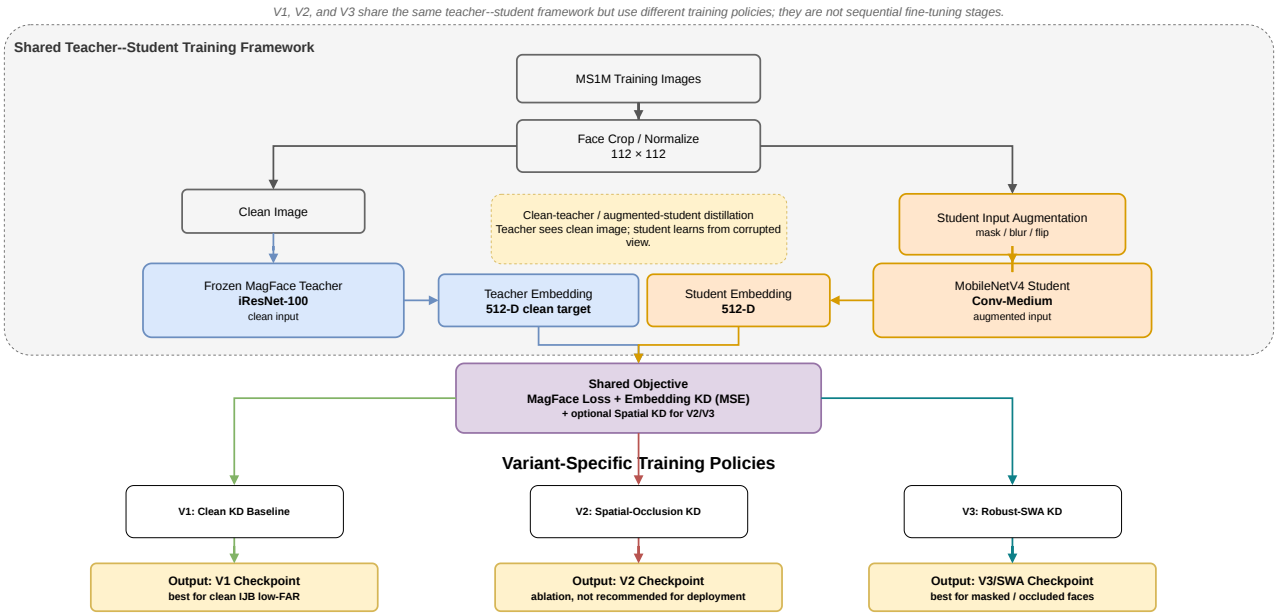


Figure 2. Training pipeline of the three independently trained variants. All variants distill a MobileNetV4 student from a frozen MagFace iResNet-100 teacher using clean teacher targets and augmented student inputs. V1 targets clean recognition, V2 evaluates aggressive spatial-occlusion KD, and V3 adds milder occlusion training with SWA for robustness.

3.3. Clean-Teacher and Augmented-Student Distillation

During training, the teacher receives the clean image, while the student receives an augmented version of the same image. This clean-teacher and augmented-student design encourages the student to produce teacher-like embeddings even when the input is degraded by masking or blur. This asymmetric input setup is shared by all training variants.

Although the repository names the models as Phase 1, Phase 2, and Phase 3, they are not sequential fine-tuning stages. They are independently trained variants of the same teacher-student framework, each trained from scratch with a different augmentation and distillation policy.

3.4. Training Variants

We evaluate three independently trained variants.

V1: Clean KD baseline. V1 focuses on clean-face recognition. It uses MagFace classification and embedding-level KD, without spatial KD. It applies only light lower-face masking after a clean warm-up period, making it suitable for clean access-control scenarios.

V2: Spatial KD with aggressive occlusion curriculum. V2 adds spatial KD and a stronger occlusion curriculum, including lower-face masking, Gaussian blur, and motion blur. However, this variant exposes a failure case: the student may receive a masked input while being forced to match the teacher's clean spatial features. This creates contradictory supervision and makes V2 mainly useful as an ablation.

V3: SWA-stabilized robust variant. V3 targets masked or partially occluded faces. It uses a milder occlusion curriculum and applies stochastic weight averaging near the end of training. SWA averages weights along the training trajectory and can improve generalization by finding flatter solutions [9]. To avoid the spatial-KD conflict, spatial feature matching is disabled when mask augmentation is active.

3.5. Runtime Face Processing

At runtime, faces are detected using a YOLO-based detector[10]. When valid five-point landmarks are available, each face is aligned to a canonical 112×112 template. Accurate face localization and landmark-based alignment are important for robust face recognition, as shown by RetinaFace [3]. If reliable landmarks are not available, the system falls back to a bounding-box crop instead of applying affine alignment with synthetic landmarks. This avoids distorted crops caused by inaccurate landmark estimates.

The detector also applies an additional cross-class non-maximum suppression step to merge duplicate detections of the same face. This reduces track fragmentation when multiple overlapping detections are produced for one person.

3.6. Tracking, Liveness, and Quality Filtering

After embedding extraction, detections are associated across frames using a tracking backend. The system supports DeepSORT, BoT-SORT, and a Hungarian matching fallback. DeepSORT improves SORT by adding an ap-

pearance association metric [22], while BoT-SORT further strengthens motion and appearance association [1].

Each tracked observation is then passed through liveness and quality filtering. Liveness detection is included because face recognition systems are vulnerable to presentation attacks such as printed photos and replayed videos. RGB-based anti-spoofing remains an important deployment challenge [13]. In our pipeline, an embedding is added to the track buffer only if the observation is classified as live and its embedding magnitude lies within the accepted quality range.

3.7. Track-Level Pooling and Retrieval

Instead of recognizing each frame independently, the system aggregates accepted embeddings over a track. Each embedding is stored with its MagFace magnitude, and the final track representation is computed by magnitude-weighted pooling followed by ℓ_2 normalization. This gives higher weight to clearer and more reliable observations.

The pooled track embedding is compared with a known identity gallery using FAISS HNSW when available, with cosine-similarity fallback otherwise. FAISS enables efficient similarity search over high-dimensional embeddings [5]. A match is accepted only if the top similarity score exceeds a threshold and the margin between the top identity and the second identity is sufficiently large. This conservative rule reduces false matches in open-set scenarios.

3.8. Unknown Identity Grouping

If a track cannot be matched to a known identity, the system groups it as an unknown identity based on embedding similarity. A track is assigned to an existing unknown group if the similarity is high enough; otherwise, a new group is created. This is related to density-based clustering methods such as DBSCAN, which can discover clusters without knowing the number of identities in advance [19].

The pipeline stores unknown-group metadata, including track IDs, frame range, prototype statistics, and sample face crops for later review. It can also optionally auto-register an unknown identity into the live index if the track satisfies minimum length and quality requirements.

3.9. Summary

Overall, the proposed method combines lightweight model distillation with a practical video recognition pipeline. During training, a MobileNetV4 student learns from a frozen MagFace teacher through MagFace classification and embedding-level KD. During deployment, the student is integrated with detection, alignment, tracking, liveness filtering, magnitude-based quality gating, track-level pooling, FAISS retrieval, and unknown identity grouping. This design addresses both edge efficiency and real-world video recognition stability.

4. Experiments

4.1. Experimental Setup

We evaluate the proposed method at two levels: face embedding verification and runtime video recognition. The first evaluation measures the discriminative quality of the learned embeddings on standard image and template-based face verification benchmarks. The second evaluation tests the trained encoder inside the full video pipeline, including detection, tracking, liveness filtering, quality gating, track-level pooling, and identity retrieval.

All student models are trained on MS1M using the same teacher-student framework[6]. The teacher is a frozen MagFace iResNet-100 model, and the student is a MobileNetV4-Conv-Medium encoder with 512-dimensional embeddings[12][16]. We compare three independently trained variants rather than sequential training stages: a clean KD baseline, a spatial-KD occlusion variant, and an SWA-stabilized robust variant[9]. Each model is trained from scratch for 40 epochs using AdamW, mixed precision, DALI RecordIO loading, a 3-epoch warm-up, and multi-step learning-rate decay at epochs 20, 30, and 36.

4.2. Training Variants

Table 1 summarizes the evaluated training variants. All variants share the same teacher, student backbone, embedding dimension, and main distillation objective, but differ in spatial KD, augmentation policy, and checkpoint stabilization.

V1 is optimized for clean-face recognition and uses only light lower-face masking after a clean warm-up period. V2 adds spatial KD and a stronger occlusion curriculum with lower-face masking, Gaussian blur, and motion blur. V3 uses a milder occlusion curriculum and applies stochastic weight averaging (SWA) near the end of training to improve robustness and checkpoint stability [9].

4.3. Evaluation Protocols

We evaluate clean image-level verification on LFW, CFP-FP, and AgeDB-30. These datasets cover general unconstrained verification, frontal-to-profile pose variation, and age variation, respectively. We report verification accuracy and the mean accuracy across the three protocols [8, 14, 18].

For template-based verification, we evaluate on IJB-B and IJB-C. These protocols are more challenging because each subject template may contain multiple still images or video frames. We report AUC and TAR at fixed FAR thresholds, with emphasis on TAR@10^{-4} because low-FAR performance is important for security-sensitive deployment [11, 21].

To measure partial-occlusion robustness, we use a synthetic lower-face masking protocol. The lower facial region is masked during evaluation, and robustness is measured by

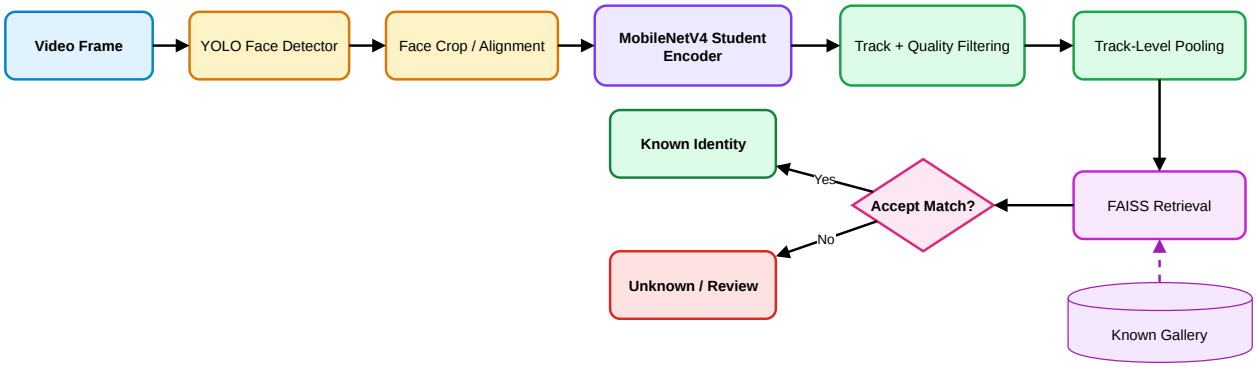


Figure 3. Inference pipeline. The system detects and aligns faces, extracts MobileNetV4 embeddings, filters reliable observations, pools embeddings at the track level, and performs FAISS retrieval to output either a known identity or an unknown label.

Table 1. Training variants evaluated in this work. Each variant is trained independently from scratch.

Variant	Spatial KD	Occlusion Policy	SWA	Purpose
V1: Clean KD	No	Light masking	No	Clean-face recognition
V2: Spatial-Occlusion KD	Yes	Aggressive masking/blur	No	Ablation study
V3: Robust-SWA KD	Yes	Milder masking/blur	Yes	Occlusion robustness

the drop in TAR compared with the clean setting. A smaller TAR drop indicates better robustness to masked or partially occluded faces.

4.4. Implementation Details

All face crops are resized to 112×112. The student receives normalized inputs in the range $[-1, 1]$, while the teacher input is remapped to $[0, 1]$ to match the MagFace checkpoint. The student is trained with a MagFace classification head and embedding-level MSE distillation from the frozen teacher. The KD weight is ramped from 5.0 to 8.0 during the first 8 epochs.

The shared training objective is:

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{MagFace}} + \lambda_{\text{kd}}\mathcal{L}_{\text{KD}} + \lambda_{\text{spatial}}\mathcal{L}_{\text{spatial}}. \quad (2)$$

For V1, $\lambda_{\text{spatial}} = 0$. For V2 and V3, spatial KD is used only when it does not conflict with active masking. In the current implementation, spatial KD is disabled during masked epochs to avoid forcing the student to match clean teacher spatial features from an occluded input.

4.5. Runtime Video Evaluation

In addition to benchmark verification, we evaluate the trained encoder inside the full runtime pipeline. The pipeline detects faces with a YOLO-based detector, aligns faces when landmarks are available, extracts embeddings with the student model, and associates detections over time using DeepSORT or BoT-SORT [1, 22]. Only observations that pass liveness and magnitude-based quality filtering are added to the track buffer. Accepted embeddings are aggregated by magnitude-weighted track-level pooling before FAISS-based retrieval [5].

For runtime evaluation, we report processed frames, mean FPS, number of detections, accepted observations, recognized observations, and match rejection reasons. When annotated video tracks are available, we additionally report open-set identification metrics such as TPIR@FPIR, FNIR, misidentification rate, and unknown false-positive identification rate.

4.6. Compared Checkpoints

We compare five checkpoints:

- **V1/latest**: clean KD checkpoint for clean-face deployment.
- **V1/best**: checkpoint selected by best mean bin-protocol performance.
- **V2/latest**: spatial-KD occlusion variant used as an ablation.
- **V3/latest**: final robust checkpoint before SWA averaging.
- **V3/SWA**: SWA-averaged checkpoint for masked or occluded deployment.

This comparison highlights the trade-off between clean verification accuracy and occlusion robustness. V1 is expected to perform best on clean benchmarks, while V3/SWA is expected to be more stable under lower-face occlusion. V2 is included as an ablation to analyze the effect of combining spatial KD with aggressive occlusion augmentation.

Table 2. Evaluation protocols and metrics.

Evaluation	Datasets / Protocols	Metrics
Clean verification	LFW, CFP-FP, AgeDB-30	Accuracy, mean accuracy
Template verification	IJB-B, IJB-C	AUC, TAR@FAR
Occlusion robustness	Masked verification protocols	TAR drop
Runtime pipeline	Recorded or live video	FPS, accepted obs., recognized obs.
Open-set video ID	Annotated video tracks	TPIR@FPIR, FNIR, MisIDR

Table 3. Clean face verification results on LFW, CFP-FP, and AgeDB-30.

Model	LFW	CFP-FP	AgeDB-30	Mean
MagFace iResNet-100	99.78	96.27	98.30	98.12
V1/latest	99.25	93.43	95.68	96.12
V1/best	99.20	**94.14**	**95.77**	**96.37**
V3/best	**99.43**	92.79	95.15	95.79
V3/latest	99.02	92.06	93.60	94.89
V3/SWA	99.00	91.54	94.00	94.85

5. Results

5.1. Clean Image Verification

Table 3 reports the verification accuracy on LFW, CFP-FP, and AgeDB-30[14, 18, 23]. The teacher model achieves the highest overall accuracy, as expected from the larger MagFace iResNet-100 backbone[12]. Among the student variants, V1 performs best on the clean bin protocols. In particular, V1/best obtains the highest mean accuracy of 96.37%, mainly due to stronger performance on CFP-FP and AgeDB-30. V3/best achieves the highest LFW accuracy among the students, but its lower CFP-FP and AgeDB-30 scores reduce its mean performance.

These results show that the clean KD variant is the most suitable checkpoint for standard clean-face verification. Although V3 is designed for robustness, its clean benchmark accuracy is lower than V1, especially on CFP-FP and AgeDB-30. This suggests a trade-off between clean verification accuracy and robustness-oriented training.

5.2. IJB Template Verification

Table 4 reports template-based verification results on IJB-B and IJB-C. We focus on TAR@10^{-4} because this operating point is more relevant for security-sensitive deployment than average accuracy alone[11, 21].

The teacher remains the strongest model on both IJB-B and IJB-C. Among the student models, V1/latest gives the best low-FAR performance, reaching 87.98% TAR@10^{-4} on IJB-B and 90.65% on IJB-C. This confirms that V1 is the preferred student checkpoint for clean or controlled deployment.

An interesting observation is that V2/latest obtains the highest AUC among the student variants but performs worse at TAR@10^{-4}. This indicates that AUC alone does not fully reflect strict low-FAR behavior. Since real access-control systems usually operate at low false-acceptance rates, TAR at strict FAR thresholds is more important for deployment.

5.3. Occlusion Robustness

Table 5 reports the drop in TAR under synthetic lower-face masking. A smaller absolute drop indicates better robustness. V3/SWA consistently reduces the degradation caused by occlusion across all evaluated protocols.

Compared with V1/latest, V3/SWA reduces the TAR drop on all occlusion benchmarks. The improvement is especially clear on AgeDB, CPLFW, and CALFW. This supports the use of V3/SWA in masked or partially occluded environments, even though it does not achieve the best clean verification accuracy[9, 14, 18, 23].

The teacher also suffers large degradation under lower-face masking, especially on CFP-FP, AgeDB, and CPLFW. This shows that a larger model is not necessarily more robust to structured occlusion. The robust student variant benefits from occlusion-aware training and SWA, making it more stable under masked-face conditions.

5.4. Overall Comparison

Table 6 summarizes the practical role of each checkpoint. V1/latest is the best choice for clean deployment because it gives the strongest student performance on IJB-B and IJB-C at strict FAR. V1/best is preferable when the goal is maximum bin-protocol accuracy. V3/SWA is the best choice when lower-face occlusion or mask usage is expected. V2/latest is kept as an ablation rather than a recommended deployment checkpoint.

Overall, the results reveal a clear trade-off. The clean KD variant achieves the best performance under clean verification protocols, while the SWA-stabilized robust variant provides the strongest resistance to lower-face occlusion. Therefore, the appropriate checkpoint should be selected based on the expected deployment environment rather than using a single model for all scenarios.

5.5. Runtime Pipeline Results

The runtime pipeline was designed to evaluate the trained encoder under realistic video conditions. It records pro-

Table 4. IJB-B and IJB-C template verification results. TAR is reported at FAR=10^{-4}.

Model	IJB-B AUC	IJB-B TAR@10^{-4}	IJB-C AUC	IJB-C TAR@10^{-4}
Teacher: MagFace iResNet-100	0.9922	93.14	0.9960	97.64
V1/latest	0.9912	**87.98**	0.9937	**90.65**
V1/best	0.9905	86.88	0.9929	89.50
V2/latest	**0.9935**	84.52	**0.9949**	86.85
V3/best	0.9916	84.77	0.9929	87.64
V3/latest	0.9917	84.78	0.9932	87.23
V3/SWA	0.9919	85.27	0.9930	87.77

Table 5. Occlusion robustness measured by TAR drop under lower-face masking. Smaller absolute drop is better.

Model	LFW	CFP-FP	AgeDB	CPLFW	CALFW
MagFace iResNet-100	-0.041	-0.520	-0.541	-0.724	-0.236
V1/latest	-0.037	-0.369	-0.462	-0.222	-0.248
V3/SWA	**-0.019**	**-0.298**	**-0.321**	**-0.148**	**-0.100**

cessed frames, detection counts, accepted observations, recognized observations, match rejection reasons, and mean FPS. However, since no standardized annotated video benchmark is used in the current evaluation, we do not report a single comparable runtime accuracy table in this section. Instead, the benchmark results above focus on the recognition quality of the trained encoders, while the runtime pipeline is used to validate practical deployment behavior with detection, tracking, liveness filtering, quality gating, pooling, and FAISS retrieval[5].

6. Discussion

6.1. Accuracy vs. Robustness Trade-off

The results demonstrate a clear trade-off between clean-face accuracy and occlusion robustness. The baseline variant (V1) achieves the best performance on clean verification, notably on IJB-B and IJB-C at FAR=10^{-4} [11, 21]. This makes V1 the optimal choice for controlled environments like access control. Conversely, the SWA-stabilized model (V3/SWA) exhibits superior resilience against lower-face masking, minimizing the True Accept Rate (TAR) drop across all masked protocols. While occlusion-oriented training slightly sacrifices clean benchmark performance, it is crucial for deployments facing frequent occlusions [11, 21]. Despite its high capacity, the teacher model suffers significant degradation under masking, proving that model size alone does not guarantee robustness.

6.2. Effectiveness of Teacher-Student Distillation

Although the MagFace iResNet-100 teacher outperforms the students on clean benchmarks due to its capacity, the MobileNetV4 student models achieve highly competitive results. This validates the efficacy of knowledge distillation for transferring the teacher's embedding structure to a lightweight model. Furthermore, the asymmetric setup, where the teacher processes clean images while the student learns from augmented ones, forces the student to map corrupted inputs to stable embeddings, effectively learning robustness without requiring the teacher itself to process corrupted inputs.

6.3. Limitations of Spatial-Occlusion KD

Variant V2 reveals that pairing spatial KD with aggressive occlusion augmentation does not improve performance. While V2 achieves high Area Under the Curve (AUC) scores on IJB protocols, its TAR at strict FAR is lower than V1 [11, 21]. This difference highlights that AUC alone is insufficient for evaluating deployment readiness in security-sensitive applications. The performance drop stems from contradictory supervision: spatial KD penalizes the student for failing to reconstruct clean spatial features from heavily masked inputs. Consequently, spatial KD should be carefully managed or disabled when processing occluded inputs, making V2 a diagnostic ablation rather than a deployment model.

6.4. The Role of Stochastic Weight Averaging

Stochastic Weight Averaging stabilizes the robustness-oriented variant. Comparing V3/latest and V3/SWA shows that SWA slightly but consistently improves low-FAR IJB performance while preserving the benefits of the occlusion curriculum [9]. Rather than maximizing clean accuracy, SWA serves as a crucial stabilization technique that yields flatter, more generalizable solutions, which is especially beneficial for masked or occluded deployment scenarios.

6.5. Importance of Track-Level Recognition

Real-world video recognition extends beyond static embedding extraction; individual frames are highly susceptible to blur, pose variations, and artifacts. The proposed runtime pipeline circumvents these issues by aggregating only reliable observations using liveness filtering, alignment, tracking, and magnitude-based quality gating [12]. Weighting clearer observations heavily during track-level pooling ensures stable and conservative identity retrieval, underscoring that practical deployment requires a holistic, end-to-end

Table 6. Recommended use cases for the evaluated checkpoints.

Checkpoint	Strength	Recommended Use
V1/latest	Best student IJB TAR@10^{-4}	Clean access-control deployment
V1/best	Best bin-protocol mean accuracy	Clean image-pair verification
V2/latest	Diagnostic spatial-KD ablation	Not recommended for deployment
V3/latest	Robust variant before averaging	Reference checkpoint
V3/SWA	Best occlusion robustness	Masked or partially occluded deployment

Table 7. Ablation factors across training variants.

Variant	Embedding KD	Spatial KD	Occlusion Strength	SWA
V1: Clean KD	Yes	No	Light	No
V2: Spatial-Occlusion KD	Yes	Yes	Aggressive	No
V3/latest	Yes	Yes	Milder	No
V3/SWA	Yes	Yes	Milder	Yes

Table 8. Clean low-FAR performance versus occlusion robustness.

Checkpoint	IJB-B TAR@10^{-4}	IJB-C TAR@10^{-4}	Occlusion Robustness
V1/latest	**87.98**	**90.65**	Medium
V3/SWA	85.27	87.77	**High**

Table 9. Effect of aggressive spatial-occlusion KD on IJB template verification.

Checkpoint	IJB-B AUC	IJB-B TAR@10^{-4}	IJB-C AUC	IJB-C TAR@10^{-4}
V1/latest	0.9912	**87.98**	0.9937	**90.65**
V2/latest	**0.9935**	84.52	**0.9949**	86.85

Table 10. Effect of SWA on the robust variant.

Checkpoint	IJB-B AUC	IJB-B TAR@10^{-4}	IJB-C AUC	IJB-C TAR@10^{-4}
V3/latest	0.9917	84.78	**0.9932**	87.23
V3/SWA	**0.9919**	**85.27**	0.9930	**87.77**

Table 11. Ablation on lower-face occlusion. Values show TAR drop; smaller absolute drop is better.

Checkpoint	LFW	CFP-FP	AgeDB	CPLFW	CALFW
Teacher	-0.041	-0.520	-0.541	-0.724	-0.236
V1/latest	-0.037	-0.369	-0.462	-0.222	-0.248
V3/SWA	**-0.019**	**-0.298**	**-0.321**	**-0.148**	**-0.100**

pipeline rather than single-frame decisions.

7. Conclusion

This work presents an edge-oriented video face recognition system based on a lightweight MobileNetV4 student distilled from a frozen MagFace iResNet-100 teacher. The proposed system combines MagFace classification, embedding-level knowledge distillation, and a complete runtime pipeline with detection, tracking, liveness filtering, magnitude-based quality gating, track-level pooling, and FAISS-based retrieval[16][15][7][5][1][22].

We evaluate three independently trained variants rather than sequential training phases. The results show that V1 is the best choice for clean-face verification, especially under strict low-FAR IJB evaluation, while V3/SWA provides better robustness under lower-face occlusion. V2 serves as an important ablation, showing that aggressive occlusion training combined with spatial KD can hurt performance when the student receives masked inputs but is forced to match clean teacher spatial features[9][21][11][5].

Overall, the study highlights a practical trade-off between clean accuracy and occlusion robustness. For controlled access-control scenarios, the clean KD variant is preferred. For environments with masks or partial occlusion, the SWA-stabilized robust variant is more suitable. Future work will focus on stronger anti-spoofing evaluation, real open-set video benchmarks, and further optimization for real-time edge deployment.

8. Limitations

This work has several limitations. First, the runtime video pipeline has not yet been evaluated on a standardized large-scale open-set video benchmark. Although the system records runtime statistics such as FPS, accepted observations, recognized observations, and match rejection reasons, a more complete evaluation with annotated video tracks is needed to measure TPIR, FPIR, FNIR, and identity switches.

Second, the anti-spoofing component is treated as a modular liveness gate rather than a fully optimized presentation attack detection system. Its effectiveness depends on the selected PAD model and the availability of real spoofing datasets. Therefore, stronger evaluation on real presentation attack protocols is required before deployment in high-security authentication scenarios.

Third, the occlusion robustness evaluation is based on synthetic lower-face masking. While this protocol is useful for controlled comparison, it does not fully represent real-world occlusions such as masks with different shapes, hands, scarves, extreme pose, motion blur, or low-resolution surveillance footage.

Finally, the three training variants are practical configuration-level ablations rather than fully isolated single-factor experiments. Since spatial KD, occlusion strength, and SWA differ across variants, future work should isolate each factor more carefully to quantify its individual contribution.

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